

MZQA

COMPREHENSIVE SINGLE-VOLUME REFERENCE

AI Analyst

System & Modeling Reference

AI Analyst is a multi-agent equity-research application: a LangGraph investment committee of LLM analysts deliberating over a deterministic valuation engine, a value/sentiment screener, and a quantitative desk that adds machine-learned return and risk forecasting and portfolio optimization built on Microsoft `qlib`, all on top of a standardized, reconciled fundamentals warehouse.

This is the comprehensive single-volume reference. It describes the system and gives the essential specification of every modeling approach — the gradient-boosted alpha model, the factor-structured risk models, the portfolio optimizers, the walk-forward backtest, the rolling factor regression, the macro-regime models, and the deterministic valuation and scoring models. Full derivations, algorithms, and hyperparameters live in the companion volumes: *System Architecture & Operations*, *Quantitative & ML Models*, and *SEC & EDINET Metrics Mapping & Reconciliation*.

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Part I System

1 Overview and architecture

AI Analyst turns a question — “*what is this stock worth, and what should I expect from it?*” — into a structured, evidence-anchored answer. It has three qualitative entry modes (single-stock committee, group/industry relative-value verdict, and an idea scanner) and a quantitative *Quant* desk.

Layer	Technology
Backend	FastAPI (async), Uvicorn, port 8027; blocking work on worker threads
Committee engine	LangGraph StateGraph; custom LLM-calling nodes
Quant layer	qlib (LightGBM, scikit-learn, cvxpy) — lean in-process embed
Frontend	Next.js 14 / React 18 / Tailwind, port 3027 (REST only)
Database	read-only PostgreSQL xbrl_sec warehouse (schema sec)
LLM layer	5 providers (DeepSeek, OpenAI, Anthropic, Moonshot, Gemini), per request

Three backend packages carry the logic: `ai_analyst/committee` (the tribunal graph, valuation, and evidence), `api` (REST routers and the `api/quant` modeling layer), and `xbrl_sec` (data access, the macro-cycle models, and the LLM client). The browser is a thin client; all financial logic and every LLM call live in the backend, whose only persistent dependency is the shared warehouse. The service topology, the committee graph, and setup are documented in full in the *System Architecture & Operations* volume.

2 The data warehouse and standardized layer

A pre-built, read-only warehouse is the single source of truth; the application never writes to it. The fundamentals it serves are not raw filings but a *standardized* layer: raw XBRL facts from SEC EDGAR (US) and EDINET (JP) are mapped onto one governed vocabulary of standardized line items, derived where filers omit subtotals, checked against accounting identities, turned into a metric library, and reconciled back to their source concepts and filings. Fundamentals are aligned point-in-time — a value with fiscal-period end p becomes usable only at $\text{monthend}(p + 90\text{d})$ — so a signal never uses data that was not yet public. Key tables: prices (`fact_prices_{us,jp,etf}`), fundamentals (`fact_metrics_{us,jp}`, `fact_fundamentals_std_{us,jp}`), factors (`fact_fama_french`, `fact_factor_loadings`), macro regime (`fact_cycle_state_monthly`, `fact_equity_factor_ic_regime`), text/sentiment (`fact_mda_sections_{us,jp}`), and 13F ownership. The mapping, derivation, metrics, and reconciliation are specified in the *SEC & EDINET Metrics Mapping & Reconciliation* volume.

3 The investment committee

A LangGraph state machine over an `InvestmentCommitteeState`. It splits into a *prepare* phase (a deterministic data gate → the financial-analysis engine → evidence nodes: news/macro, 13F, a data-quality agent, and qlib quant signals) and a *debate* phase (Advocate, Challenger, Auditor and sector specialists → Lead → memo). The prepare phase is provider-independent, so several LLM providers can debate the same prepared evidence concurrently. Specialists include a Quantitative Factor Analyst that interprets the quant-desk signals. The output is a probability-weighted fair value, a triangulation range, scenarios, an evidence bundle, and a written memo — a valuation with its reasoning, not a single buy/sell label. A deterministic completeness gate is a hard stop; the accounting-identity data-quality check is advisory by default and blocks only under an opt-in strict mode.

4 Screener and idea generation

The screener filters the US/JP universe on fundamentals (deterministic SQL) and offers a natural-language screen (LLM → filters). The *value + sentiment* scanner produces a 0–100 interest score (§14.1); the group ranking produces a deterministic relative-value composite (§14.2). The prompt screen fans out across every keyed provider and unions the results; the group verdict runs a single structured narrative over the deterministic ranking.

5 The quant desk

A lean, in-process integration of `qlib`: a cross-sectional alpha model (return prediction, §7), a factor-structured risk model (§8), dual portfolio optimization (native *and* `qlib` backends, §9), and a walk-forward out-of-sample backtest with Fama–French benchmarking (§10). It is exposed via a `/api/quant/*` router and a *Quant* tab, and fed into the committee as an evidence node. The engineering of the embed is documented in the *qlib Integration* volume.

Part II Modeling reference

This part gives the essential specification of each model — its object, its estimator, and its key equations. Full derivations, algorithms, and shipped hyperparameters are in the *Quantitative & ML Models* volume.

6 Notation

$\mathbf{w} \in \mathbb{R}^N$ portfolio weights; $\boldsymbol{\mu} \in \mathbb{R}^N$ expected (annualized) returns; $\Sigma \in \mathbb{R}^{N \times N}$ return covariance; r_f risk-free rate; $\mathbf{1}$ the ones vector; z a cross-sectional z -score. Time subscripts t index months unless stated; annualization uses 252 trading days (daily) or 12 (monthly).

7 Cross-sectional alpha model

`api/quant/qlib_alpha.py`, on `qlib`'s `LGBModel`. On a monthly, cross-sectionally z -scored, point-in-time fundamentals panel, a gradient-boosted tree ensemble $F_M(\mathbf{z}) = \sum_{m=1}^M \eta h_m(\mathbf{z})$ predicts the forward 1m/3m return. Each stage fits the residual (the negative gradient of the squared-error loss), with LightGBM's leaf-wise histogram split gain

$$\text{Gain} = \frac{1}{2} \left[\frac{(\sum_{I_L} g_i)^2}{\sum_{I_L} h_i + \lambda} + \frac{(\sum_{I_R} g_i)^2}{\sum_{I_R} h_i + \lambda} - \frac{(\sum_I g_i)^2}{\sum_I h_i + \lambda} \right] - \gamma. \quad (1)$$

Predictive power is the per-month information coefficient $\text{IC}_t = \text{Corr}(\hat{y}_{\cdot,t}, y_{\cdot,t})$ and its rank analogue RankIC_t , summarized by $\text{ICIR} = \overline{\text{IC}} / \text{std}(\text{IC})$. Performance is quoted from the walk-forward out-of-sample path (§10), which is materially weaker than the in-sample test segment.

8 Factor-structured risk models

`api/quant/qlib_risk.py`, `api/quant/risk.py`. Returns are driven by K latent factors, giving a low-rank-plus-diagonal covariance

$$\Sigma = F \Sigma_B F^\top + \text{diag}(\mathbf{d}), \quad (2)$$

with F the factor exposures (PCA/factor analysis), Σ_B the factor-return covariance, and $\mathbf{d}$ the specific variance — a stable shrinkage of the noisy sample covariance, annualized $\times 252$. Two alternatives are selectable: Ledoit–Wolf shrinkage toward a constant-correlation target $\Sigma^* = \delta F + (1 - \delta)S$, and a Fama–French *fundamental* covariance $\Sigma = B F_{\text{ann}} B^\top + D$ built from the named-factor loadings of §11. Forward volatility per name is $\sigma_i = \sqrt{\Sigma_{ii}}$.

9 Portfolio optimization

`api/quant/qlib_optimize.py` dispatches at runtime to a native optimizer or any `qlib` optimizer; both consume the same $\boldsymbol{\mu}, \Sigma$. The native optimizer maximizes a multi-term mean–variance utility

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2} \lambda \mathbf{w}^\top \Sigma \mathbf{w} - \gamma_f \|S \odot a \odot (B^\top \mathbf{w} - \mathbf{t})\|_2^2 - \gamma_{\text{to}} \|\mathbf{w} - \mathbf{w}_0\|_1 - \gamma_c \|\mathbf{w}\|_2^2 \quad (3)$$

subject to full investment, per-name bounds, a gross cap, sector caps, an annualized volatility cap, and factor caps/ranges — solved by SLSQP with an analytic gradient, or as a `cvxpy` mixed-integer program when a position-count (cardinality) constraint is set. The `qlib` backends provide GMV (min variance), MVO, risk parity, inverse-vol, and enhanced indexing (benchmark-relative tracking-error, consuming the $(F, \Sigma_B, \mathbf{d})$ decomposition). For a solution $\mathbf{w}^*$ the desk reports $\mathbb{E}[r]_{\text{ann}} = \mathbf{w}^{*\top} \boldsymbol{\mu}$, $\sigma_{\text{ann}} = \sqrt{\mathbf{w}^{*\top} \Sigma \mathbf{w}^*}$, and the Sharpe ratio.

10 Walk-forward out-of-sample backtest

`api/quant/qlib_backtest.py`. For each test month, a booster trained on all prior months ranks that month's names (investable universe, winsorized forward returns), and the equal-weight top- k (optionally long/short) book realizes the next return — out-of-sample by construction. The monthly series is scored for annualized return, volatility, Sharpe, Sortino, max drawdown, and hit rate, benchmarked against the Fama–French market (excess, tracking error, information ratio, β), and regressed on the FF five factors plus momentum to separate alpha from factor exposure:

$$r_t - r_{f,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t, \quad t_\alpha = \hat{\alpha} / \text{se}(\hat{\alpha}). \quad (4)$$

11 Rolling Fama–French factor regression

`xbrl_sec/sec/sources/factor_model.py`. Per ticker, over a rolling 252-day window, excess returns are regressed on the FF factors (FF3/FF4/FF5/FF6), $r_{i,t} - r_{f,t} = \alpha_i + \sum_k \beta_{i,k} f_{k,t} + \varepsilon_{i,t}$, estimated with the robust Huber loss (down-weighting outlier days) and Newey–West HAC standard errors (robust to autocorrelation/heteroskedasticity). Stored diagnostics include adjusted R^2 , residual volatility $\hat{\sigma}_\varepsilon \sqrt{252}$ (the idiosyncratic D of the FF covariance), Durbin–Watson, the condition number, and F -statistics. These loadings feed the WACC (§13.1) and the fundamental covariance.

12 Macro-cycle regime models

`xbrl_sec/sec/cycle/`. A PCA/dynamic-factor baseline compresses a monthly multimodal feature set (macro, market, fundamental) into latent cycle factors $\mathbf{x}_t$. Two regime models run on top. A Gaussian Hidden Markov Model ($\mathbf{x}_t \mid s_t=k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$, transition A , prior π) fit by Baum–Welch yields the smoothed state posterior $\gamma_t(k)$; an optional temporal multimodal VAE maximizes the evidence lower bound $\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})] - \beta \text{KL}(q_\phi \parallel p)$ and thresholds its reconstruction-stress percentile into phases. States are mapped to business-cycle phases (contraction / late-cycle / mid- / early-expansion), NBER-calibrated for the US. The regime is linked to forecasting through the regime-conditioned factor IC:

$$\text{IC}_{\text{regime}}(m) = \rho_{\text{Spearman}}(x_{m,i}, r_i^{\text{fwd}} \mid \text{regime}), \quad (5)$$

stored to `fact_equity_factor_ic_regime` and used to re-weight the scanner's factor families (§14.1).

13 Valuation models

`ai_analyst/`. Deterministic, on a 7-year explicit horizon. A DCF projects revenue, EBIT, NOPAT, and free cash flow, with Gordon terminal value and the enterprise/equity bridge

$$\text{TV} = \frac{\text{FCF}_7(1 + g_\infty)}{\text{WACC} - g_\infty}, \quad \text{EV} = \sum_{i=1}^7 \frac{\text{FCF}_i}{(1 + \text{WACC})^i} + \frac{\text{TV}}{(1 + \text{WACC})^7}. \quad (6)$$

The cost of capital (§13.1) builds cost of equity from the FF betas and long-run premia, $R_e^{\text{CAPM}} = r_f + \beta_{\text{Mkt}} \text{ERP}$, blends a synthetic-rating cost of debt, and weights them by market-value capital structure. A reverse DCF solves for the growth or margin the market is pricing in; sum-of-the-parts values each segment on its own growth/margin and a segment WACC; scenario weighting and triangulation combine the DCF, SOTP, and peer-multiple ranges into a probability-weighted fair value and a football field.

13.1 Cost of capital

$\text{WACC} = \frac{E}{V} R_e + \frac{D}{V} R_d(1 - \tau)$, with E market cap, D total financial debt, R_e the FF/CAPM cost of equity, and $R_d = r_f$ + spread from a Damodaran-style synthetic rating on interest coverage.

14 Scoring and ranking

14.1 Value + sentiment interest score

`screener_agent.py`. A 0–100 score blending a normalized qlib alpha v_α (dominant term, $w_\alpha \approx 0.40$) with a fundamental composite whose value/growth budget is re-split by the current regime’s factor IC:

$$\text{score} = w_\alpha v_\alpha + (1 - w_\alpha) \sum_k w_k v_k, \quad \text{interest} = 100 \cdot \text{score}. \quad (7)$$

With no model or IC it reduces exactly to the fixed-weight composite.

14.2 Group relative-value composite

`committee/group.py`. A cross-sectional z -score $z_{k,i}$ per metric (P/E, EV/EBITDA, P/B, FCF yield, dividend, growth, margins) and a signed weight w_k (negative for cheaper-is-better multiples) give $\text{composite}_i = \sum_k w_k z_{k,i}$, ordered descending into attractive/fair/expensive terciles with a per-name audit trail that sums exactly to the composite. Loss-making multiples are excluded from their population. MD&A management tone (an LLM read in $[-1, 1]$) feeds the interest score’s tone term.

Part III Reference

15 How the models connect

The *valuation* path (DCF/WACC/SOTP/reverse/triangulation) answers what a business is worth; the *quant* path ($\alpha \rightarrow \mu$, structured covariance $\rightarrow \Sigma$, optimizer $\rightarrow \mathbf{w}$, backtest) answers what to expect and how to size it; the *macro-cycle* path (HMM/VAE $\rightarrow$ regime-conditioned IC) conditions the scoring on the regime; and the *committee* fuses the deterministic evidence and the quant signals into a reasoned, auditable verdict. The full model map is in the *Quantitative & ML Models* volume.

16 REST API (active surface)

Base `http://127.0.0.1:8027`, under `/api: meta, screener (+/ai), screener/agent/value-sentiment, ai/committee (+/prepare, /debate, /iterate), ai/committee/group, prices, kpis, company, fx, llm`, and the quant desk `quant/{backends,alpha,risk,optimize,backtest}`.

17 Running

Backend (venv Python, from `backend/` so `qlib` resolves): `uvicorn api.main:app -port 8027`; frontend `npm run dev` (port 3027); or `start_committee.bat`. `qlib` is built from the bundled clone (no PyPI wheel for Python 3.13) and installed non-editable so `import qlib` resolves from any directory. The quant desk pre-warms the alpha cross-section and the walk-forward backtest at startup, so the first user request is served from cache.

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